SD1	Understand and use a probability density function, $\mathrm{f}(x)$, for a continuous distribution and understand the differences between discrete and continuous distributions. Understand and use distributions of random variables that are part discrete and part continuous.

	Content
SD2	Find the probability of an observation lying in a specified interval.

	Content
SD3	Find the median and quartiles for a given probability density function, $\mathrm{f}(x)$.

	Content
SD4	Find the mean, variance and standard deviation for a given pdf, $\mathrm{f}(x)$. Know the formulae $\mathrm{E}(X) = \int x\mathrm{f}(x)\mathrm{d}x$, $\mathrm{E}\left(X^2\right) = \int x^2\mathrm{f}(x)\mathrm{d}x$, $\mathrm{Var}(X) = \mathrm{E}\left(X^2\right) - \left(\mathrm{E}(X)\right)^2$

	Content
SD5	Understand the expectation and variance of linear functions of CRVs and know the formulae: $\mathrm{E}(aX + b) = a\mathrm{E}(X) + b$ and $\mathrm{Var}(aX + b) = a^2\mathrm{Var}\left(X\right)$ Know the formula $\mathrm{E}\left(\mathrm{g}(X)\right) = \int \mathrm{g}(x)\mathrm{f}(x)\mathrm{d}x$ Find the mean, variance and standard deviation of functions of a continuous random variable such as $\mathrm{E}\left(5X^3\right), \mathrm{E}\left(18X^{-3}\right), \mathrm{Var}\left(6X^{-1}\right)$

	Content
SD6	Understand and use a cumulative distribution function, $\mathrm{F}(x)$. Know the relationship between $\mathrm{f}(x)$ and $\mathrm{F}(x)$. $\mathrm{F}(x) = \int_{-\infty}^{x} \mathrm{f}(t)\mathrm{d}t$ and $\mathrm{f}(x) = \frac{\mathrm{d}}{\mathrm{d}x}\mathrm{F}(x)$

	$\text{f}(x) = \begin{cases} b - a & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$ Know the conditions for the rectangular distribution to be used as a model. Calculate probabilities from a rectangular distribution. Know proofs of mean, variance and standard deviation for a rectangular distribution.

	Content
SD8	Know that if X and Y are independent (discrete or continuous) random variables then $\text{E}(X + Y) = \text{E}(X) + \text{E}(Y)$ and $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$